
Vincenzo Ruotolo

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Software Engineer

Software Engineer with experience in consultancies, start-ups and as a contractor. Deep expertise in implementing robotics architecture, core algorithm modules, autonomous systems, scalable cloud app and ETL pipelines for telemetry and test data. Secondary strengths in embedded systems, ML/AI, DevOps and automated testing. Proven track record delivering production-grade software for automotive ADAS, cloud-native APIs, and robotic platforms. Experienced in leading and coordinating cross-functional agile teams.

WORK EXPERIENCE

Autonomous Robotic Systems

Senior software developer (Confidential Client) Remote • 06/2024 - Present

Designed a **scalable, modular robotics architecture** for multi-platform autonomy (UGV / UAV / UUV) prioritizing component reuse, versioned interfaces and safe upgrade paths.

- Develop and deploy core algorithm modules (SLAM, GNC and Sensors) in C++/ROS2
- Implement embedded firmware, hardware drivers and system integration
- Lead and coordinate a cross-functional agile team of 8 engineers

Luxoft (DXC)

Software automation developer (Ford) Guadalajara, Mexico • 08/2022 - 05/2024

Engineered Python-based automation testing framework for Ford's BlueCruise (ADAS L2+), boosting validation coverage and reducing testing time.

- Built a **Python automation & telemetry pipeline** for BlueCruise (ADAS L2+), combining pytest orchestration with ETL to ingest, normalize and replay vehicle telemetry for deterministic HIL validation.
- Implemented **CAN / UDS / TCP / UDP** adapters for trace capture, injection and end-to-end replay
- Developed unit/integration/smoke/sanity suites and Python harnesses that shifted many checks off HIL into CI, reducing regression time

Software developer (BMW) Turin, Italy • 08/2021 - 07/2022

C++ software developer for Adaptive AUTOSAR application layer.

- Implemented C++ modules for **Adaptive AUTOSAR (ARA::COM)**; designed efficient serialization/deserialization for ECU data flows
- Created C++ unit/integration tests and Python bindings for automated validation
- Enforced MISRA-inspired coding practices and static analysis in CI

Cosmico

Software developer

Remote • 05/2020 - 07/2021

SKILLS

OS: Ubuntu, Kali, Windows

Programming: C++, C, Python, Matlab/Simulink, Bash

Data: Pandas, SQL (PostgreSQL • SQLAlchemy), NoSQL (MongoDB • PyMongo • Redis • Pydantic), FastAPI, Kafka, Airflow, Spark

Robotics: ROS2, micro-ROS, Gazebo

Embedded: FreeRTOS, IoT, RPi

ML/AI: Scikit-learn, Pytorch, OpenCV

DevOps: Git, GitHub/GitLab, Ansible, Docker, Agile/Scrum, CI/CD, Terraform, AWS

Testing: TDD, pytest/unittest, gtest/gmock

Build tools: Make, CMake, Bazel

Languages: Italian, English, Spanish

EDUCATION

Master's Degree in Mechatronic Engineer

Politecnico di Torino

Thesis: Optimal Trajectory Generation via MPC for a Four-Wheel Steering UGV

Bachelor's Degree in Aerospace Engineer

Politecnico di Torino

Thesis: Thales Alenia Space internship

Built a **cloud-native backend** for automated technical interview evaluation using **FastAPI** and **SQLAlchemy**, implementing candidate management, pre-formed assessments and result aggregation.

- Built cloud-native backend (FastAPI + SQLAlchemy) for automated interview evaluation; implemented candidate management and assessment orchestration
- Implemented ETL pipelines to ingest candidate submissions (code, logs, recordings), normalize and persist artifacts (S3), and produce queryable datasets for analytics and recruiter dashboards, processing 1,500+ assessments per month with 98% uptime
- Integrated ML/AI inference for automated scoring (code quality, tests, plagiarism detection) and designed retraining pipelines using labeled outcomes and feedback loops

PROJECTS

European Space Agency - Space Resources Challenge 10/2021 - 04/2023
DRAFT PoliTO

The challenge aims to develop an autonomous system of UGV and UAV, collaborating in a unknown space environment. Led the robotics architecture, core module development, integration, and DevOps.

Leonardo Drone Contest 10/2019 - 10/2021
DRAFT PoliTO

The contest aims to develop autonomous drones in contest of GPS-denied condition. Designed the high-level robotics architecture, led component integration and simulation.